

HECTOR GENARO PACHECO

Software Engineer | Backend • Cloud Infrastructure • DevOps

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PROFESSIONAL SUMMARY

Backend and cloud-focused software engineer with experience across standalone cloud architectures and collaborative team development in shared codebases. Hands-on experience building containerized REST & GraphQL microservices, relational PostgreSQL/Prisma schemas, AWS ECS Fargate deployments, zero-secret runtime configuration (AWS Secrets Manager/KMS), Jira-driven ticket workflows, Git/GitLab version control, and automated testing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell	Backend & APIs: Node.js, Express, REST APIs, GraphQL, Apollo Server, FastAPI, Zod, RBAC
Cloud & Infra: AWS (ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC, IAM), Vercel	Containers & OS: Docker, Multi-Stage Builds, Container Registries (ECR), Linux, systemd
Databases: PostgreSQL, Prisma ORM, Neon, Supabase, Relational Modeling, Indexing	DevOps & Testing: Git, GitLab, GitHub, Jira, Docker CLI, Vitest, Integration Testing, Build Verification
Architecture: Microservices, API Contract Design, Data Isolation, Deterministic Math	Security & Auth: JWT Authentication, HTTP-only Cookies, Secrets Injection, Least Privilege

FEATURED SOFTWARE & SYSTEMS ENGINEERING PROJECTS

CalTrack — *Industrial Calibration SaaS & API Platform* [LIVE APP ↗](#) Node.js • Express • TypeScript • PostgreSQL • Prisma • Docker • AWS ECS Fargate • ECR

- [GITHUB ↗](#)
- Architected and deployed containerized Node/Express REST API on AWS ECS Fargate with Amazon RDS PostgreSQL in isolated VPC subnets.
 - Built multi-stage Docker pipeline compiling shared TypeScript packages, generating Prisma client, running automated tests, and building linux/amd64 images.
 - Configured Amazon ECR image hosting and injected runtime environment credentials securely using AWS Secrets Manager and AWS KMS.
 - Enforced role-based access control (RBAC), database foreign-key constraints, and historical snapshot preservation for compliance auditing.

EstateOS — *Real Estate Investment Intelligence Platform* [CASE STUDY ↗](#) TypeScript • Node.js • Next.js 16 • PostgreSQL • Prisma ORM • Vitest • Zod

- Engineered deterministic backend underwriting engine and data models in TypeScript and Prisma, processing mortgage amortization, capital stacks, and LTV/CLTV.
- Built server-controlled PropertyEvidenceSnapshot domain in PostgreSQL, isolating trusted municipal assessor/tax data from untrusted client inputs.
- Established rigorous automated test harness achieving 804 passing tests across 59 test files using Vitest, validating debt math invariants and schema validation.

Procurement Tracking System — *Enterprise GraphQL Platform* Node.js • Express 5 • GraphQL • Apollo Server 4 • Prisma ORM 6 • PostgreSQL • JWT

- [LIVE APP ↗](#) [GITHUB ↗](#)
- Designed and deployed a type-safe GraphQL purchasing API on Render with PostgreSQL on Neon, modeling complex purchase-order and multi-stage receiving state transitions.
 - Implemented Apollo Server 4 query/mutation resolvers with JWT authentication, HTTP-only session cookies, and multi-role RBAC resolver permission guards.

Additional Repositories & Deployments: **Interactive Photo Booth** — Linux systemd Daemon (Raspberry Pi 5/Python) [CASE STUDY ↗](#) • **OC Water Features** — Production Web Platform (Next.js/Supabase/Vercel) [LIVE SITE ↗](#) • **Pi Arcade OS** — Embedded Linux System (Python/GPIO) [SITE ↗](#) [GITHUB ↗](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Software Engineering Contributor — *Nonprofit serving underserved communities* Collaborative Contribution

- Collaborated on an existing nonprofit codebase using Jira-driven ticket workflows, Git/GitLab version control, API integration investigation, local environment configuration, and shared build and test workflows.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* Field Engineering Experience

- Maintained high-availability industrial control infrastructure, diagnosing electrical loops, PLC I/O wiring, relays, and field instrumentation across 480VAC, 120VAC, and 24VDC systems with systematic root-cause isolation.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026)